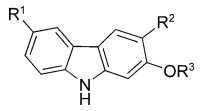
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General Au-Catalyzed Benzannulation Towards Naturally Occurring Carbazole Alkaloids from Methoxypropadiene

Wangqing Kong, Chunling Fu, and Shengming Ma*[a]

A variety of structurally diverse carbazole alkaloids have been isolated from different natural sources over the past years.[1] The genus *Murraya*, trees growing in southern Asia, represents the richest source of carbazole alkaloids from terrestrial plants, particularly for 2-oxygenated tricyclic carbazole alkaloids (Scheme 1). Some of these compounds show useful biological activities. For example, in 2000, Boyd et al.[2] developed a bioassay-guided fractionation of the extract leading to the isolation of a new carbazole alkaloid, siamenol, which shows interesting anti-HIV activity (EC_{50} = 2.6 μg mL^{-1}), reaching 50–60% maximum protection in the 2,3-bis-(2-methoxy-4-nitro-5-sulfophenyl)-2H-tetrazolium-5-carboxanilide (XTT) assay from *Murraya siamensis* collected in Thailand. 3-Formyl-7-hydroxycarbazole was isolated from the root bark of *Murraya euchrestifolia* (Scheme 1). Extracts of the leaves and bark of *Murraya* have been used as a folk medicine for analgesia as well as local anesthesia, for treatment of eczema, rheumatism, dropsy, abdominal pain, stomach-ache, toothache, diarrhoea, oedema, thrombosis and stasis of blood, and as an expectorant, anti-convulsant, anodyne, and detoxification agent.[3] Clauszoline-K, clauszoline-L, clausine-M and clausine-N were isolated from the stem bark of the Chinese medicinal plant. *Clausena excavata* is a wild plant that has been claimed to be useful in folk medicine as a detoxification agent and for the treatment of snake bites.[4]



R^1	R^2	R^3	Name	biological activities
Me	Prenyl	H	Siamenol	anti-HIV (EC_{50} = 2.6 μg/mL)
COOH	H	Me	Clausine-N	
COOMe	H	Me	Clausine-C	Potential components in folk medicines
CHO	H	Me	Clauszoline-K	
COOMe	H	H	Clausine-M	
CHO	H	H	3-Formyl-7-hydroxycarbazole	

Scheme 1. Naturally occurring 2-oxygenated tricyclic carbazole alkaloids.

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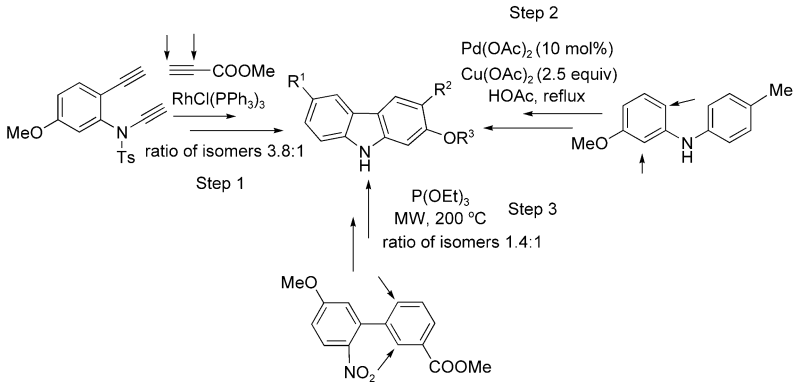
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In 2002, Witulski et al.[5] reported a creative approach employing a rhodium-catalyzed [2+2+2] cyclotrimerization of a diyne and an alkyne as the key step to clausine-C starting from 2-iodo-5-methoxyaniline. However, the regioselectivity of this step is rather poor (the ratio of isomers is 3.8:1; step 1 in Scheme 2). In 2006, Knölker et al.[6] reported a remarkable approach for the general synthesis of clauszoline-K, 3-formyl-7-hydroxycarbazole, clausine-M, clausine-N, clausine-C, and siamenol by utilization of the palladium-catalyzed intermolecular C–N bond formation between 3-methoxyaniline and 1-bromo-4-methylbenzene, followed by intramolecular C–C bond formation of the coupling product, 3-methoxy-*N*-*p*-tolylaniline, as the key step (step 2 in Scheme 2). However, there should be a serious issue of selectivity for this type of cyclization of unsymmetric 3-methoxy-*N*-*p*-tolylaniline according to the related previous reports,[7] although it was not discussed in this report.[6] In addition, large amounts of oxidants, such as $Cu(OAc)_2$ (2.5 equiv), DDQ (4.2 equiv), and MnO_2 (26 equiv), are required to complete the synthesis.

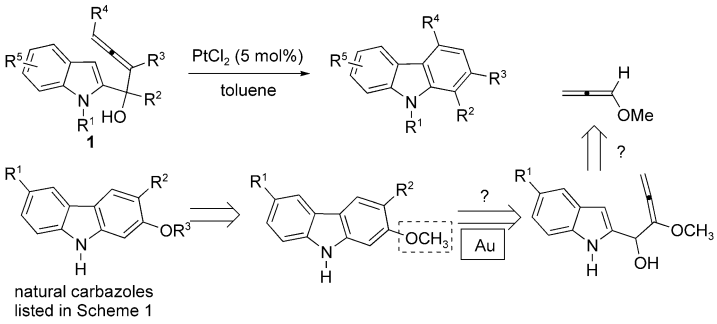
Furthermore, in the first total synthesis of siamenol,[6] the coupling of 3-bromo-6-methyl-9*H*-carbazol-2-ol required 9.5 equiv of prenyl bromide and 12 equiv of $Ni(cod)_2$ (cod = 1,5-cyclooctadiene) for six days to yield siamenol in 47% yield with a low purity. Thus, the pure siamenol had to be prepared by preparative HPLC, which makes this method much less attractive. In 2008, Kuethe et al.[8] reported a route employing Suzuki–Miyaura cross-coupling of a 2-nitrobenzenediazonium salt with 2-(methoxycarbonyl)phenylboronic acid to give methyl 3-(2-nitro-3-methoxyphenyl)-benzoate, which was followed by reductive cyclization to afford clausine-C and clausine-N. However, it is obvious that the atom economy is very low and these required reagents are not readily available. In addition, the selectivity is also very low in the step of reductive cyclic amination (the ratio of isomers is 1.4:1; step 3 in Scheme 2). The regioselectivity issue is obviously deterring scientists from easily obtaining the pure compounds. Thus, for the purpose of systematic biological screening to discover lead compounds, highly selective and diverse routes with atom economy to these naturally occurring carbazoles from readily available starting materials are highly desirable.

In 2009, we developed a $PtCl_2$-catalyzed synthesis of carbazoles from 1-indoyl-2,3-allenols (Scheme 3).[9] However, it is difficult to introduce functionalities to the skeleton by using alkyl-substituted allenols. We envisioned that using

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Scheme 2. Reported strategies for the targets listed in Scheme 1.

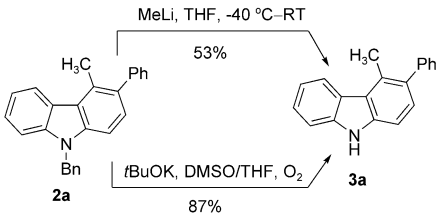


Scheme 3. Strategy for the synthesis of carbazoles listed in Scheme 1.

readily available methoxypropadiene as the starting point would solve this problem by incorporating the methoxy group in the to-be-formed benzene ring for further manipulation to prepare these 2-functionalized naturally occurring carbazoles (Scheme 3). The challenges here will be the efficient route to the required allenols and the influence of the methoxy group on the reactivity caused by the selectivity of the coordination of the two C=C bonds with the metal. In this paper, we wish to present such a general strategy to these naturally occurring carbazoles under Au catalysis.

It is noted that in all the carbazoles prepared in this laboratory, there is no *N*-substituent. Thus, we must identify a protecting group for the N–H functionality that could be removed after the formation of the carbazole. This task was not trivial, since most of the classical protecting groups, such as *N*-Tosyl and *N*-Boc, failed to produce the carbazoles due to the fact that the electron-withdrawing groups on the nitrogen atom make the indole electron-deficient, and therefore nonreactive towards the current cyclization. Finally we observed that the benzyl group could be introduced into carbazole **2a** to afford the N-unprotected carbazole **3a** in 53% yield with MeLi.[10] By using potassium *tert*-butoxide/DMSO and O_2,[11]the yield for the deprotection is much higher (87%), which was applied in further study (Scheme 4).

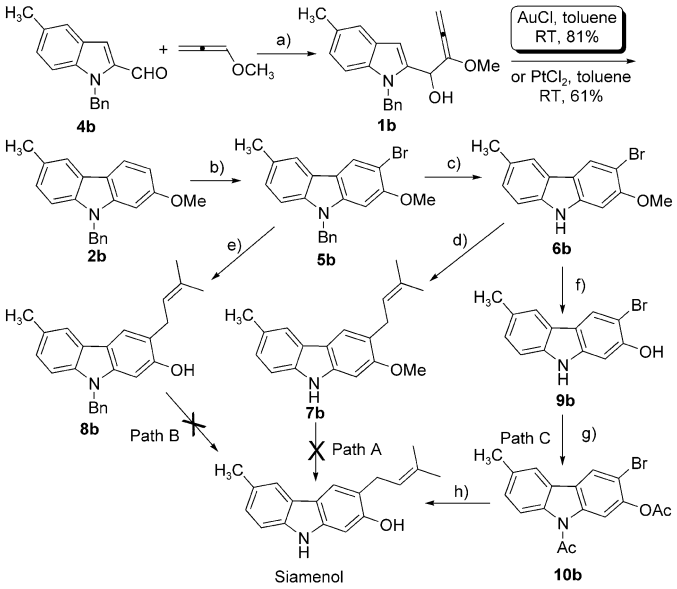
Interestingly, we synthesized 2,3-allenol **1b** conveniently from the readily available 1-methoxypropa-1,2-diene: the reaction of 1-benzyl-5-methyl-indole-2-carbaldehyde **4b** with 1-methoxypropa-1,2-dienyl lithium, in situ generated from the treatment of methoxypropadiene[12] and *n*BuLi, afforded the required methoxy-substituted allenol **1b** in 70% yield.[13] Gladly, the $PtCl_2$-catalyzed cyclization of **1b** under the standard reaction conditions did afford carbzaole **2b**,[9] albeit in a low yield (61%) probably due to the presence of methoxy group. Interestingly, further screening led to the observation that **2b** was formed exclusively in a much higher yield (81%) with AuCl as the catalyst. Electrophilic bromination of carbazole **2b** afforded 3-bromocarbazole **5b** in 88% yield as the only product. Deprotection of the *N*-benzyl group using potassium *tert*-butoxide/DMSO and O_2 afforded **6b** in 80% yield (Scheme 5). Next, we started to establish an effective coupling protocol: Suzuki cross-coupling reaction of **5b** with pinacol prenylboronate[14] afforded **7b** in 72% yield. However, in survey of many demethylation conditions, acidic conditions utilizing Lewis acids, such as BBr_3, afforded a complicated mixture of several products, probably due to the intrinsic problem of oxygen anion attack at the allylic double bond. Many reported protocols were tried for this purpose: lithium chloride in refluxing DMF,[15] sodium ethyl mercaptide in refluxing DMF,[16] and PhSH in the presence of a catalytic amount of K_2CO_3 in refluxing *N*-methyl-2-pyrrolidone (NMP),[17] but all led to decomposition of compound **7b** (path A, Scheme 5). On the other hand, Suzuki cross-coupling reaction of **5b** with pinacol prenylboronate and demethylation with sodium ethyl mercaptide in refluxing DMF afforded **8b** in 60% yield in two steps. Disappointingly, deprotection of the *N*-benzyl group of **8b** with potassium-*tert*-butoxide (*t*BuOK) in DMSO under O_2, 2,3-dichloro-5,6-dicyano-1,4-benzoquinone (DDQ) in CH_2Cl_2[18] or MeLi in THF[19] all failed to generate siamenol (path B, Scheme 5). Finally, demethylation of **6b** with BBr_3 in CH_2Cl_2 afforded **9b** in 88% yield. Initially, we tried the direct Pd^0-catalyzed Suzuki cross-coupling reaction of **9b** with pinacol prenylboronate but this failed, probably due to the effect of the free *ortho*-hydroxyl group. When *tert*-butyldimethylsilyl (TBS) ether was used for the protection of the free hydroxyl group, the Suzuki coupling reaction did not occur. Eventually, we were pleased to note that the protec-



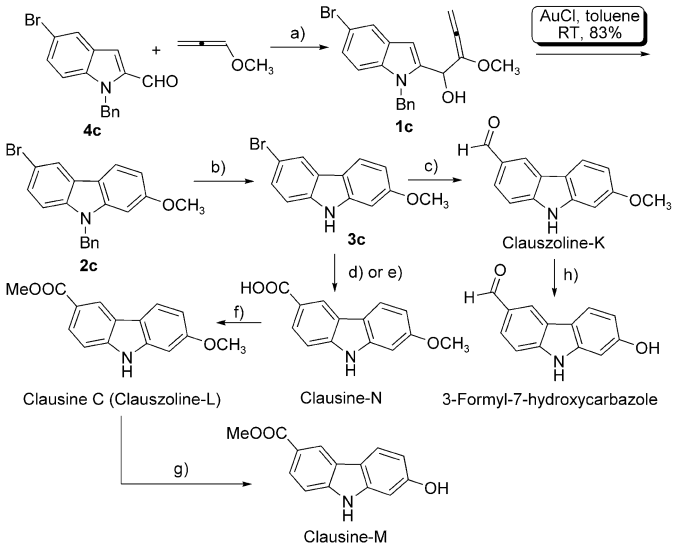
Scheme 4. Debenzylation of the carbazole **2a**.

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Scheme 5. Synthesis of siamenol. a) *n*BuLi, −40 °C, Et_2O, 70 %; b) NBS, CCl_4, reflux, 30 min, 88%; c) *t*BuOK, DMSO, O_2 (1 atm), RT, 5 h, 80%; d) pinacol prenylboronate (2 equiv), $Pd(PPh_3)_4$ (5 mol %), NaOH (20 equiv), toluene, 90 °C, 1 h, 72%; e) (1) pinacol prenylboronate (2 equiv), $Pd(PPh_3)_4$ (5 mol %), NaOH (20 equiv), toluene, 90 °C, 1 h; (2) EtSNa (6 equiv), DMF, reflux, 60% (two steps); f) BBr_3 (2 equiv), CH_2Cl_2, −78 °C–RT, 88 %; g) DMAP, Et_3N, Et_2O, Ac_2O, 11 h, 90%; h) pinacol prenyl-boronate (2 equiv), $Pd(PPh_3)_4$ (5 mol %), NaOH (20 equiv), toluene, 90 °C, 1 h; NaOH (20 equiv), CH_3OH, 90 °C, 1 h, 66 %.



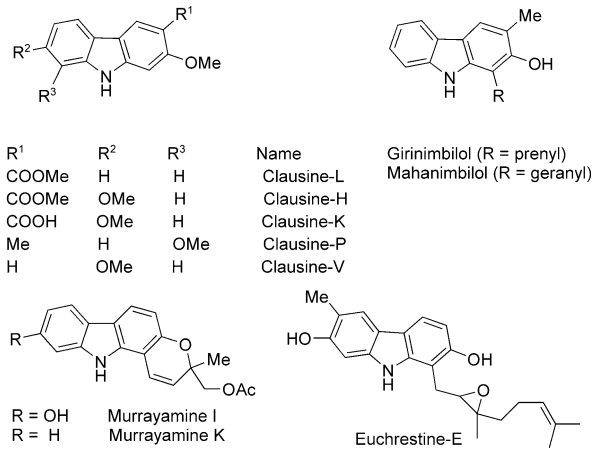
Scheme 6. Efficient synthesis of targets listed in Scheme 1 except siamenol. a) *n*BuLi, −40 °C, Et_2O, 68 %; b) *t*BuOK, DMSO/THF, O_2 (1 atm), RT, 72 %; c) (1) KH, THF, RT, (2) *t*BuLi, −40 °C, DMF, 54%; d) (1) KH, THF, RT, (2) *t*BuLi, −78 °C, CO_2, 84%; e) (1) KH, THF, RT, *t*BuLi, −78 °C, ClCOOMe; (2) NaOH, CH_3OH, 90 °C, 42 % (two steps); f) MeI, $NaHCO_3$, DMF, 95 %; g) BBr_3 (4.3 equiv), CH_2Cl_2, −78 °C to −20 °C, 60 %; h) BBr_3 (4.5 equiv), CH_2Cl_2, −78 °C to −20 °C, 54 %.

tion of the OH group as acetate, followed by coupling and instant deprotection in a one-pot reaction afforded siamenol in 66 % yield (path C, Scheme 5).

Luckily, a similar approach may also be used for the synthesis of allenol **1c**. The AuCl-catalyzed cyclization of **1c** is also very smooth under the standard reaction conditions to afford exclusively carbazole **2c** in 83 % yield. The deprotection of the *N*-benzyl group with potassium *tert*-butoxide in DMSO under O_2 (Scheme 4) afforded 6-bromo-2-methoxy-9*H*-carbazole **3c** in 72 % yield. Starting from **3c**, clauszoline-K was synthesized in 54 % yield by the subsequent treatment with potassium hydride, *t*BuLi, and DMF to install the aldehyde functionality;[20] clausine-N was synthesized in 84 % yield by using the Li–Br exchange reaction with *t*BuLi and the subsequent carboxylation with CO_2; methylation of this carboxylic acid functionality in clausine-N with MeI gave clauszoline-L in 95 % yield with $NaHCO_3$ as the base. Removal of the methyl group in clauszoline-K and clauszoline-L by treatment with BBr_3 in CH_2Cl_2 with a modified procedure led to 3-formyl-7-hydroxycarbazole and clausine-M, respectively (Scheme 6).[6,21]

The spectroscopic data of our synthetic compounds, that is, siamenol (7 steps with 21 % overall yield), clauszoline-K (4 steps with 22 % overall yield), clauszoline-L (5 steps with 32 % overall yield), 3-formyl-7-hydroxycarbazole (5 steps with 11 % overall yield), clausine-M (6 steps with 19 % overall yield), and clausine-N (4 steps with 34 % overall yield), from the readily available methoxypropadiene are in full agreement with those reported for the corresponding natural products (for detailed information, see the Supporting Information).

In summary, the approach reported herein has solved the issue of regioselectivity in all the previous reports.[5,6,8] It is obvious that these routes may be easily used for large scale syntheses of these naturally occurring carbazole alkaloids or diversified highly selective synthesis of their derivatives due to the substituent-loading capability of both the indole moiety and allenol functionality for further biological screening. In addition, this strategy should be general to the synthesis of other naturally occurring compounds shown in Scheme 7, based on **2b**- or **5b**-type products.[22] This chemistry will be of high interest for organic chemists and medici-



Scheme 7. Other naturally occurring 2-oxygenated carbazole alkaloids.

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nal chemists. Further studies in this area including biological screening are being carried out in our laboratory.

Experimental Section

Typical procedure for the preparation of 9-benzyl-6-methyl-2-methoxy-9*H*-carbazole (2b): $PtCl_2$ (2.8 mg, 0.01 mmol), **1b** (64.0 mg, 0.20 mmol), and toluene (2 mL) were added sequentially to a dry Schlenk tube under a nitrogen atmosphere. After continuous stirring for 24 h at room temperature, the reaction was complete, as monitored by TLC. Evaporation and column chromatography on silica gel (petroleum ether/ethyl acetate = 20 L) afforded **2b** as a solid (36.7 mg, 61 %). M.p. 129–130 °C (*n*-hexane/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 7.94$ (d, $J = 8.4$ Hz, 1H, ArH), 7.82 (s, 1H, ArH), 7.30–7.07 (m, 7H, ArH), 6.86–6.75 (m, 2H, ArH), 5.41 (s, 2H, $ArCH_2$), 3.84 (s, 3H, $ArOCH_3$), 2.51 ppm (s, 3H, $ArCH_3$); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 159.0$, 142.2, 139.0, 137.2, 128.7, 128.6, 127.3, 126.3, 125.7, 123.3, 121.0, 119.6, 116.7, 108.3, 107.2, 93.2, 55.6, 46.5, 21.4 ppm; IR (KBr): $\tilde{\nu} = 2994$, 2869, 1635, 1610, 1498, 1492, 1452, 1342, 1296, 1260, 1196, 1168, 1142, 1058 cm^{-1}; MS (EI) (70 eV): *m*/*z* (%): 302 $[M+1]^+$ (23.12), 301 $[M]^+$ (100); elemental analysis calcd (%) for $C_{21}H_{19}NO$: C, 83.69; H, 6.35; N, 4.65; found: C, 83.57, H, 6.50; N, 4.61.

The reaction of AuCl (21.4 mg, 0.092 mmol) and **1b** (604.5 mg, 2.0 mmol) in toluene (10 mL) at room temperature for 8 h afforded **2b** (481.6 mg, 81 %; petroleum ether/ethyl acetate = 40 L).

Acknowledgements

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Keywords: alkaloids · annulation · carbazoles · methoxypropadienes · natural products · synthetic methods

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